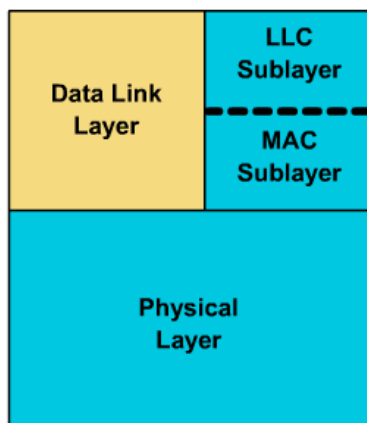


03 Medium Access Control Sublayer

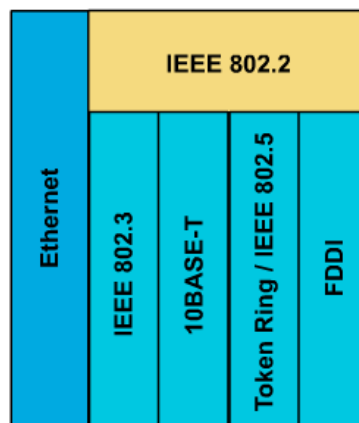
03.01 Channel assignment problems, Multiple access protocols

OSI and Various LAN Standards

OSI Layers



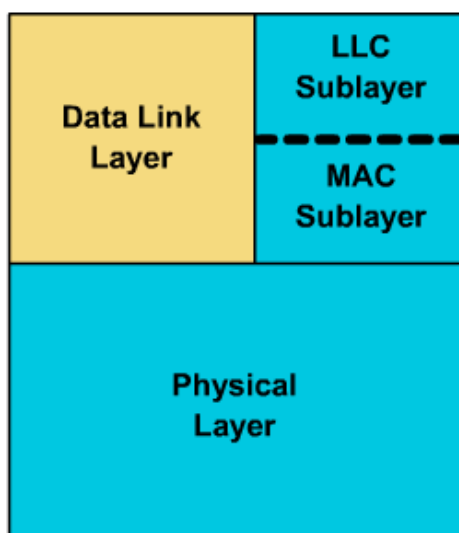
LAN Specification



IEEE and DLL

- The **I**nstitute of **E**lectrical and **E**lectronic **E**ngineers.
- LAN standards:
 - **802.1d:** **Spanning tree.**
 - **802.2:** **LLC.**
 - **802.3:** **MAC ~ Ethernet.**
 - **802.5:** **MAC ~ Token ring.**
 - **802.11:** **Wireless LAN.**

IEEE and LAN standards



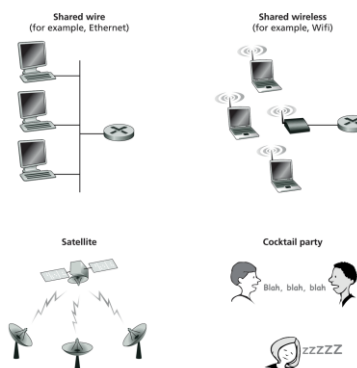
- **Logical Link Control (LLC):**
Transitions up to the network layer.
- **Media Access Control (MAC):**
Transitions down to media.

TWO Sublayers

- **LLC** serves to communicate upward to Network layer, **independent** of the specific LAN technology used and Upper layer.
- **MAC** serves to access and communicate downward to the technology-specific Physical layer.

The Channel Allocation Problem

- Broadcast may present more transmitters/receivers on the same shared channel
- These nodes need to be coordinated → the multiple access problem → multiple access protocols
- Protocols can be classified in:
 - Channel partitioning protocol
 - Random access protocols
 - Taking-turn protocols



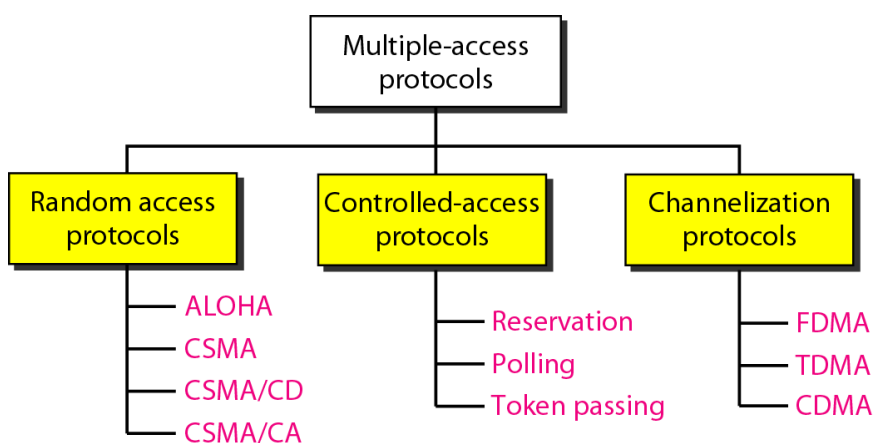
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♦ Various multiple access channels

The Channel Allocation Problem

- All nodes may send frames → possible *collisions*
→ frames are no more readable → need re-transmission → bandwidth waste if too frequent
- Ideal characteristics of a multiple access protocol (on a channel of R bit/sec):
 - When only 1 node is sending data, it should have R bps
 - When M nodes need to send data, they can do it at R/M bps (on average)
 - It should be decentralized (to avoid system failures in case of master node faults)
 - It should be as simple as possible to implement

Taxonomy of MAC protocols



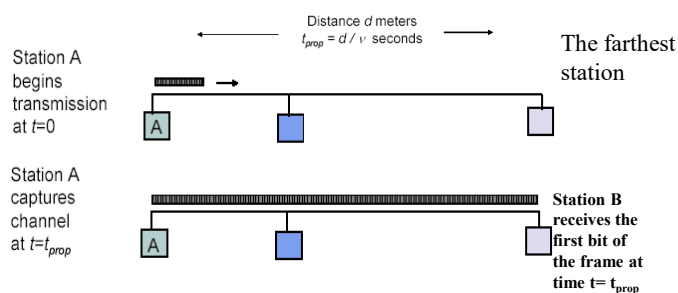
Random access protocols

Random Access (or contention) Protocols:

- No station is superior to another station and none is assigned the control over another.
- A station with a frame to be transmitted **can use the link directly based** on a procedure defined by the protocol to make a decision on whether or not to send.

Maximum Propagation Delay

- **Maximum propagation delay**(t_{prop}): time it takes for a bit of a frame to travel between the **two most widely** separated stations.

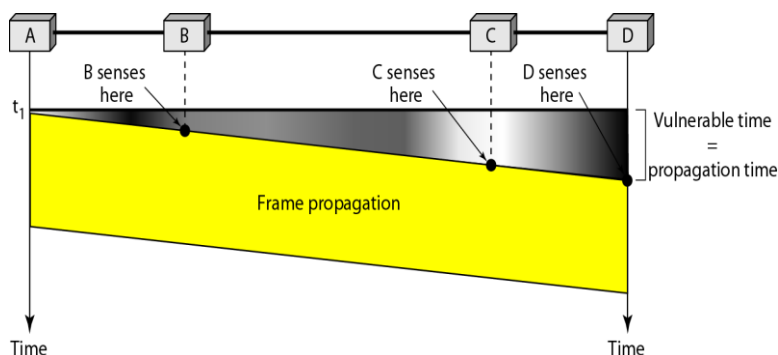


Random access protocols

- CSMA (carrier sense multiple access):
 - In ALOHA nodes decide what to do independently from other nodes
 - If a protocol *listens* to the channel to sense if a transmission (carrier) is being done is called *carrier sense protocol*
 - CSMA → “if someone is transmitting, wait until it has finished”
 - CSMA/CD (collision detection) → “if someone starts to transmit with you, stop your transmission and try again later” (used in Ethernet)

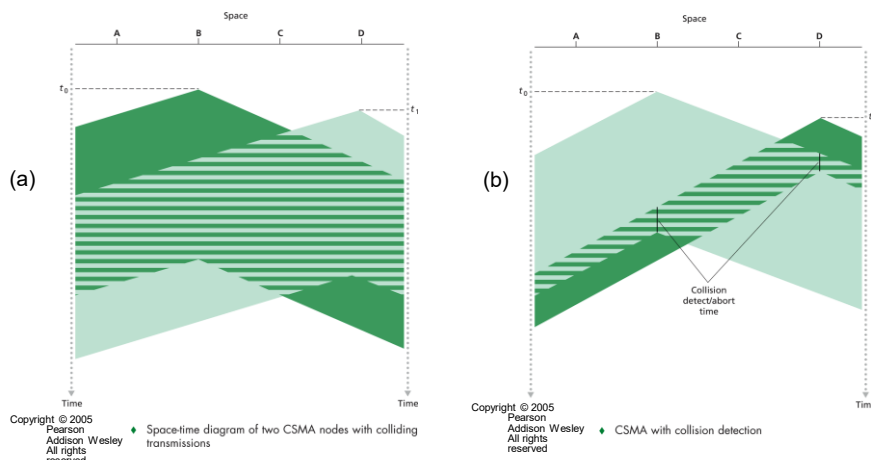
Random Access - CSMA

- Vulnerable time for CSMA is the maximum propagation time
- The longer the propagation delay, the worse the performance of the protocol because of the above case.



CSMA and CSMA/CD

- If the transmitter is able to detect collisions it can rapidly stop transmissions hence saving time and bandwidth (compare figs. (a) and (b))



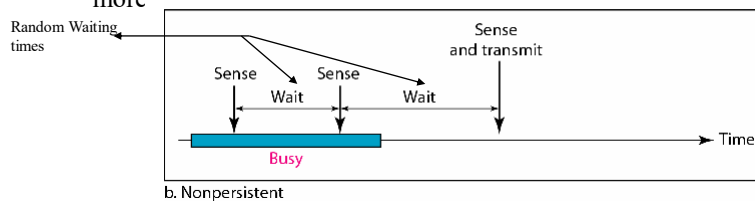
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Nonpersistent CSMA

- A station with frames to be sent, should sense the medium
 - If medium is idle, **transmit**; otherwise, go to 2
 - If medium is busy, (**backoff**) wait a **random amount of time** and repeat 1
- Non-persistent Stations are **deferential (respect others)**
- Performance:
 - Random delays reduces probability of collisions because two stations with data to be transmitted will wait for different amount of times.
 - Bandwidth is **wasted** if waiting time (backoff) is large because medium will remain idle following end of transmission even if one or more



b. Nonpersistent

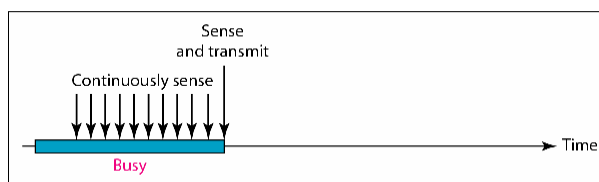
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1-persistent CSMA

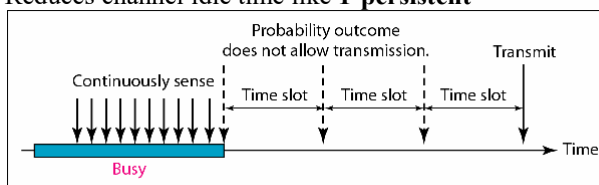
- To avoid idle channel time, 1-persistent protocol used
- Station wishing to transmit listens to the medium:
 - If medium idle, **transmit** immediately;
 - If medium busy, **continuously listen** until medium becomes idle; then transmit immediately with probability 1
- Performance:
 - 1-persistent stations are **selfish**
 - If two or more stations becomes ready at the same time, **collision guaranteed**



a. 1-persistent

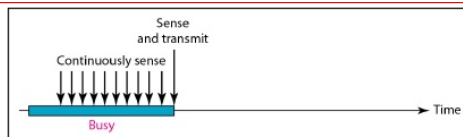
P-persistent CSMA

- Time is divided to slots where each Time unit (slot) typically equals **maximum propagation delay**
- Station wishing to transmit listens to the medium:
 - If medium idle
 - transmit with probability (p), OR
 - wait **one time unit (slot)** with probability ($1 - p$), then repeat 1.
 - If medium busy, **continuously listen until idle** and repeat step 1
- Performance
 - Reduces the possibility of collisions like **nonpersistent**
 - Reduces channel idle time like **1-persistent**

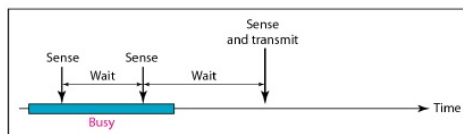


c. p-persistent

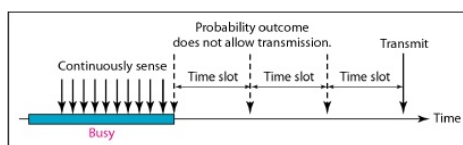
Behavior of Three Persistence methods - recap



a. 1-persistent

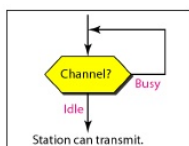


b. Nonpersistent

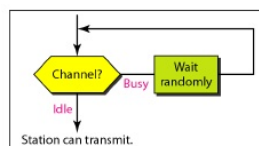


c. p-persistent

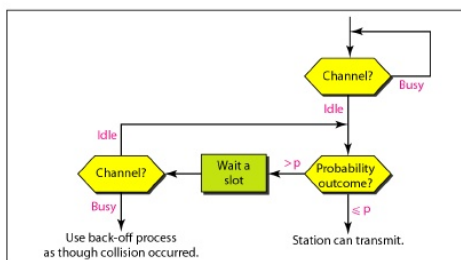
Persistence Methods



a. 1-persistent

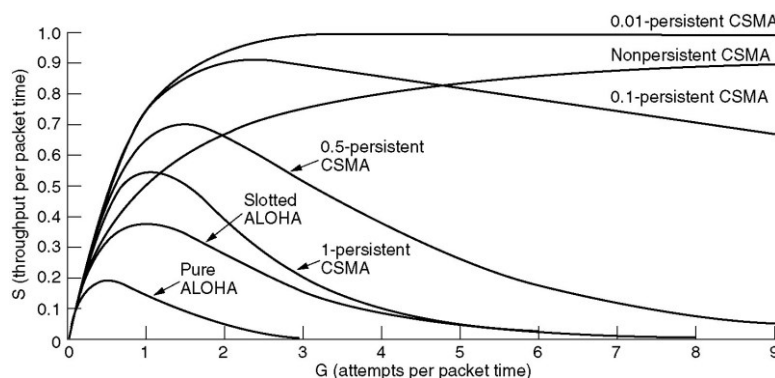


b. Nonpersistent



c. p-persistent

Access Methods Comparison



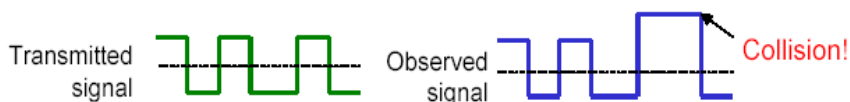
Comparison of the channel utilization versus load for various random access protocols.

CSMA/CD

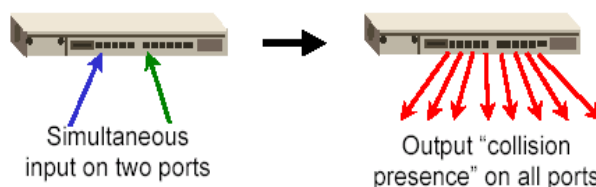
- **CSMA (all previous methods) has an inefficiency:**
 - If a collision has occurred, the channel is **unstable** until colliding packets have **been fully transmitted**
- **CSMA/CD (Carrier Sense Multiple Access with Collision Detection) overcomes this as follows:**
 - While transmitting, the sender is **listening to medium** for collisions.
 - Sender **stops transmission** if collision has occurred **reducing channel wastage**.
- CSMA/CD is Widely used for **bus topology LANs** (IEEE 802.3, **Ethernet**).

Detect a collision

Transceiver: A node monitors the media while transmitting. If the observed power is more than transmitted power + attenuated reflection of its own signal, it indicates a collision.



Hub: if input occurs simultaneously on two ports, it indicates a collision. Hub sends a collision presence signal on all ports.



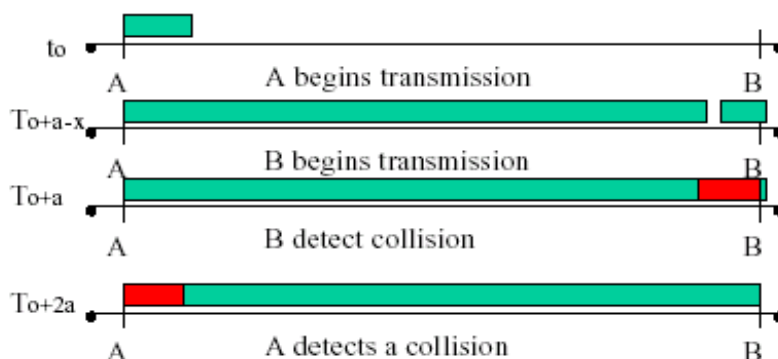
CSMA/CD Protocol

- Use one of the CSMA persistence algorithm (*non-persistent*, *1-persistent*, *p-persistent*) for transmission
- If a collision is detected by a station during its transmission then it should do the following:
 - **Abort transmission** and
 - **Transmit a *jam signal* (48 bit)** to notify other stations of collision so that they will **discard the transmitted frame** also to make sure that the collision signal will stay until detected by the furthest station
 - After sending the *jam signal*, **backoff (wait) for a random** amount of time, then
 - Transmit the frame again

CSMA/CD

- **Question:** How long does it take to detect a collision?
- **Answer:** *In the worst case, twice the maximum propagation delay of the medium*

Note: a = maximum propagation delay



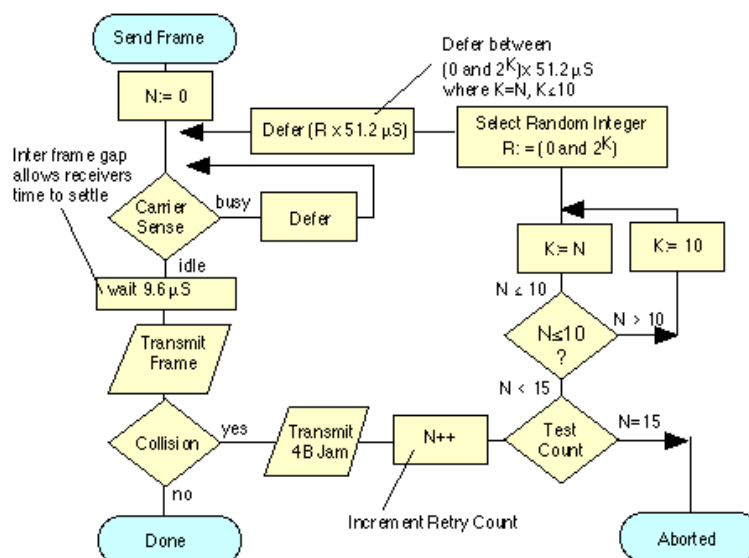
CSMA/CD: Restrictions

- Packet **transmission time** should be **at least** as long as the time needed to detect a collision ($2 * \text{maximum propagation delay} + \text{jam sequence transmission time}$)
- Otherwise, CSMA/CD does not have an advantage over CSMA

Exponential Backoff Algorithm

- Ethernet uses the **exponential backoff algorithms** to determine **the best duration of the random waiting period after the collision happens**
- Algorithm:
 - Set “**slot time**” equal to $2 \times \text{maximum propagation delay} + \text{Jam sequence transmission time}$ (= 51.2 usec for Ethernet **10-Mbps LAN**)
 - After K^{th} collision, select a random number (R) between 0 and $2^k - 1$ and **wait** for a period equal to $(R \times \text{slot time})$ then **retransmit** when the medium is **idle**, **for example**:
 - After first collision ($K=1$), select a number (R) between 0 and $2^1 - 1$ {0,1} and wait for a period equal to $R \times \text{slot times}$ (Wait for a period 0 usec or 1×51.2 usec) then retransmit when the medium is idle
 - Do not increase random number range, if $K=10$
 - Maximum interval {0 – 1023}
 - Give up after 16 unsuccessful attempts and report failure to higher layers

Exponential Backoff Algorithm



Exponential Backoff Algorithm

- Reduces the chance of two waiting stations picking the same random waiting time
- When network traffic is light, it results in **minimum** waiting time before transmission
- As congestion increases (traffic is high), collisions increase, stations backoff by **larger amounts** to reduce the probability of collision.
- Exponential Back off algorithm gives **last-in, first-out effect**
 - Stations with **no or few collisions** will have the chance to transmit before stations that have waited longer because of their previous unsuccessful transmission attempts.

Throughput in CSMA-CD

- Depends on Persistence Method used
- For 1-persistence max throughput is for $G=1$ (near 50%)
- For Non-persistence method and for $3 < G < 8$, max throughput is near 90%
- G is the average number of frames generated per slot

Random Access Protocols

Performance

- Simple and easy to implement
- Decentralized (no central device that can fail and bring down the entire system)
- In low-traffic, packet transfer has low-delay
- However, limited throughput and in heavier traffic, packet delay has no limit.
- In some cases, a station may never have a chance to transfer its packet. (**unfair protocol**)
- A node that has frames to be transmitted can **transmit continuously** at the **full rate of channel (R)** if it is the **only node with frames**
- If (M) nodes want to transmit, many collisions can occur and the rate for each node will **not be on average R/M**